
Adaptive Color Correction for Colorblindness in Human Computer Interaction

By

Gazi Rafat Al Kafi - 011 223 0876

Sanjida Hossain - 011 223 1039

Ridwanoor Rahman Rafi - 011 223 0801

MD Zobaer Islam - 011 223 0184

Mohaiminul Islam - 011 222 030

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UNITED INTERNATIONAL UNIVERSITY

Abstract

Colour Vision Deficiency affects around 300 million people, roughly 8% of men and 0.5% of women. For them, digital interfaces built around colour alone are often just broken. Existing accessibility solutions, such as operating system-level colour filters and daltonization algorithms, apply static, one-size-fits-all transformations designed primarily for severe dichromacy. But most CVD users have anomalous trichromacy, with severity that varies significantly from person to person. This mismatch means that many users receive either no meaningful benefit or a distorted visual experience that is not tailored to their actual perception.

This thesis presents a **Adaptive Color Correction (ACC)**, a software system that provides personalised, real time colour correction for CVD users across digital applications. The proposed system integrates three components: (1) an online CVD assessment module based on a digitised hue ordering test to estimate the user's CVD type and severity parameter, $\delta \in [0, 1]$; (2) a severity conditioned colour transformation model trained on simulated CVD image pairs using a model such as the Machado-Oliveira-Fernandes method; and (3) a cross application screen layer rendering pipeline that runs as a background application and applies per pixel colour correction in real time. In contrast to prior work limited to static image processing or specialist AR hardware, ACC operates directly on the screen output of any application and requires neither a clinical diagnosis nor specialised hardware.

Evaluated against fixed filter daltonization baselines using CIEDE2000 (ΔE) colour difference metrics and an Ishihara plate recognition study, the system produced statistically significant improvements in colour discrimination, with effect sizes consistent with the broader CVD correction literature.

Keywords: Colour Vision Deficiency, CVD, Daltonization, Personalised Recolouring, Real Time Screen Accessibility, Machine Learning, Anomalous Trichromacy, Digital Accessibility.

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List of Abbreviations

Abbreviation	Full Form
AR	Augmented Reality
API	Application Programming Interface
CIEDE	Commission Internationale de l'Éclairage Delta E
CNN	Convolutional Neural Network
CVD	Colour Vision Deficiency
D-15	Farnsworth D-15 Colour Arrangement Test
DFD	Data Flow Diagram
DL	Deep Learning
DVAL	Dynamic Visual Accessibility Layer
FM100	Farnsworth-Munsell 100-Hue Test
FPS	Frames Per Second
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
HMD	Head-Mounted Display
HSV	Hue-Saturation-Value (colour space)
LMS	Long-Medium-Short (cone cell colour space)
ML	Machine Learning
MLP	Multilayer Perceptron
OS	Operating System
RGB	Red-Green-Blue (colour space)
SUS	System Usability Scale
SVM	Support Vector Machine
UIU	United International University
UI	User Interface
URP	Universal Render Pipeline
WCAG	Web Content Accessibility Guidelines
XYZ	CIE 1931 XYZ Colour Space
ΔE	Colour difference / colour deviation metric
δ	CVD severity parameter (range 0–1)

Chapter 1

Introduction

1.1 Project Overview

Colour is one of the primary channels through which digital interfaces communicate information, from red-highlighted error messages to graphs, charts, data visualisations, and navigation elements. For users with Colour Vision Deficiency (CVD), however, this information channel is partially or wholly unavailable. CVD is a common hereditary condition affecting approximately 300 million people globally, roughly 8% of men and 0.5% of women worldwide [1]. It arises from anomalies in one or more of the three types of cone photoreceptors in the retina (L, M, and S cones), resulting in varying degrees of difficulty distinguishing specific hues.

Operating systems like Windows, macOS, iOS, and Android currently handle digital accessibility for CVD with fixed colour filters that apply the same static transformation to screen output, regardless of a user's specific type or severity of CVD. This one-size-fits-all approach fundamentally misaligns with user needs. Most CVD users have an anomalous trichromacy, in which all three cone types are present but one has shifted sensitivity. The degree of impairment varies widely from person to person [2]. A fixed filter optimised for severe dichromacy often distorts colour more than it helps when applied to a mild anomalous trichromat.

This project addresses that gap with a Dynamic Visual Accessibility Layer (DVAL). A software system that estimates each user's CVD type and severity through an interactive online assessment, applies a severity-conditioned color transformation tailored to that individual's perceptual profile, and delivers the correction in real time across any screen content through a browser extension. The system does not require a prior clinical diagnosis and is not built for any single application. It works as a universal cross-application screen layer.

1.2 Motivation

Despite three decades of research and widespread deployment of CVD accessibility features in consumer operating systems and applications, user surveys consistently show dissatisfaction with existing solutions [3]. A scoping review of 113 academic papers on CVD accessible design found that existing tools predominantly focus on

simulation or high contrast filtering, with very few addressing personalized, degree adaptive correction [4]. Studies involving CVD users themselves found that many users view blanket recolouring filters negatively, often believing that developers did not understand the actual problem they were trying to solve [3].

The technical literature reflects the same gap. Zhu et al. [5] introduced a degree adaptable recolouring algorithm that accounts for anomalous trichromacy through a severity parameter δ , but their optimization based method requires approximately 30 seconds per image and is therefore unsuitable for real time use. Qin et al. [2] demonstrated personalized real time CVD correction in an Augmented Reality setting with statistically significant improvements (Cohen's $d > 1.4$), but their system requires a Meta Quest 3 AR headset and is limited to AR content. No existing system delivers personalized, severity conditioned, real time colour correction as a software layer for general purpose screen use.

From the user perspective, Sajek et al. [6] tested 128 HTML colours with 18 CVD participants and confirmed that colour type and severity significantly interact with how web content is perceived, reinforcing the need for a personalized approach over a universal contrast recommendation to genuinely improve usability.

1.3 Objectives

This project designs, builds, and evaluates a personalized, real-time screen colour correction system for users with colour vision deficiency (CVD) that doesn't depend on a clinical diagnosis. Specifically, it aims to:

- O1. Develop a CVD profiling module** that estimates a user's CVD type (protan, deutan, or tritan) and severity $\delta \in [0, 1]$ using a digital interactive hue-ordering test, requiring no specialist hardware or clinical administration.
- O2. Design and train a severity conditioned colour correction model** that maps input pixel colours to corrected output colours based on the user's CVD profile, improving perceptual distinguishability while preserving naturalness.
- O3. Implement a cross application real time screen rendering layer** as a background process that intercepts and transforms screen content on the fly, at a frame rate suitable for everyday use.
- O4. Evaluate the system** quantitatively using colour difference metrics (ΔE / CIEDE2000) against established baselines, and qualitatively through a user study with CVD participants, measuring Ishihara plate recognition before and after correction.

- O5. Compare the personalized approach with fixed filter approaches** to establish whether degree-adaptive correction improves on standard OS accessibility filters.

1.4 Methodology

The project follows a three phase methodology.

Phase 1 Offline Model Training: a dataset of image pairs is generated by simulating CVD perception on natural images using the Machado-Oliveira-Fernandes physiologically based model [7], parameterized by CVD type and severity δ . A severity conditioned colour transformation model (initially a parametrized 3×3 matrix in LMS colour space conditioned on $(type, \delta)$) is trained to minimize the perceptual colour difference ΔE between confused colour pairs.

Phase 2 Online User Profiling: An interactive assessment module presents the user with a digitized hue ordering task derived from the Farnsworth-Munsell methodology. The user's arrangement errors are used to classify CVD type and estimate severity δ , producing a personal CVD profile without requiring clinical instruments.

Phase 3 Real-Time Screen Correction: The trained model, conditioned on the user's profile, is deployed as a browser extension using the Web Extensions API. A pixel shader applies per pixel colour transformation to all rendered content in real time, targeting a performance of less than 16ms per frame (60 FPS).

A detailed description of each phase appears in Chapter 3.

1.5 Project Outcome

The expected outcomes of this project are:

1. A functional **application** implementing the Dynamic Visual Accessibility Layer, demonstrable in real time.
2. A **trained colour correction model** conditioned on CVD type and severity, packaged for client side inference.
3. A **CVD profiling interface** that produces a user profile from an interactive hue ordering assessment in under two minutes.
4. **Evaluation results** comparing the personalized DVAL against fixed filter baselines on both objective colour metrics and subjective user study measures.
5. A **research contribution** establishing the novelty of the end-to-end personalised screen layer pipeline relative to the existing literature.

1.6 Organization of the Report

The remainder of this thesis is organized as follows:

Chapter 2 Background provides preliminary definitions of CVD types and the relevant colour science, reviews existing similar applications and related academic research across the four main clusters of prior work, and identifies the research gaps this project addresses.

Chapter 3 Project Design details the requirement analysis, system architecture, data flow, UI design, detailed methodology for each component, project plan, and task allocation.

Chapter 4 Implementation and Results describes the development environment, testing procedures, evaluation protocol, and the quantitative and qualitative results obtained.

Chapter 5 Standards and Design Constraints addresses software, hardware, and communication standards compliance, engineering design constraints across economic, ethical, environmental, health, social, political, and sustainability dimensions, cost analysis, and the complex engineering problem characterization.

Chapter 6 Conclusion summarizes the project's contributions, notes its limitations, and outlines directions for future work.

Chapter 2

Background

2.1 Preliminaries

2.1.1 Colour Vision and the Human Visual System

Human colour vision depends on three types of light-sensitive cone photoreceptor cells in the retina, each responsive to different wavelengths of visible light: long-wavelength (L) cones, sensitive to red; medium-wavelength (M) cones, sensitive to green; and short-wavelength (S) cones, sensitive to blue [4]. Colour perception arises from the comparative excitation levels across these three cone types, with the brain constructing colour experience from the ratios of L, M, and S responses rather than from any single channel in isolation.

2.1.2 Colour Vision Deficiency: Types and Classification

Colour vision deficiency (CVD) arises when one or more cone types are absent, non-functional, or spectrally shifted. The condition is classified along two dimensions: the affected cone type and the severity of the deficiency [4].

Dichromacy is the more severe form and results when one cone type is entirely absent. The three dichromatic subtypes are:

- *Protanopia* : missing L (red) cones; reds appear very dark or black.
- *Deutanopia* : missing M (green) cones; greens and reds appear as yellows or browns. This is the most common severe form.
- *Tritanopia* : missing S (blue) cones; blues appear greenish and yellows appear grey or violet. Rare (<0.001% of the population).

Anomalous trichromacy is less severe and results when all three cone types are present but one has an atypical spectral sensitivity, usually a wavelength shift [4]. The three subtypes are protanomaly (L cone shifted), deuteranomaly (M cone shifted), and tritanomaly (S cone shifted). Anomalous trichromacy is far more prevalent than dichromacy: deuteranomaly alone affects approximately 5% of males.

Together, red green CVD (protan and deutan types) affects approximately 8% of men and 0.5% of women globally [2], meaning that of the roughly 5.16 billion active

internet users worldwide, hundreds of millions encounter colour inaccessible digital content daily [6].

2.1.3 Colours of Confusion and Confusion Lines

People with CVD perceive certain pairs of colours as identical, despite those colours being clearly distinct to trichromatic observers. These are known as *colours of confusion*. In the CIE colour space, colours that appear identical to dichromats lie along *confusion lines*: lines that converge toward a copunctal point specific to each CVD type. For protanopia, reds, greens, and oranges all appear as similar yellows; for deuteranopia, a similar but not identical pattern occurs; for tritanopia, blues and yellows are the affected axis [4].

2.1.4 Colour Spaces Used in CVD Research

Several colour spaces are relevant to this project:

- **RGB**: the standard display colour space. Not perceptually uniform.
- **LMS**: models the three cone types directly. Essential for physiologically accurate CVD simulation; the Machado model [7] operates in this space.
- **CIELab**: a perceptually uniform colour space. Colour difference E is measured here, with ΔE is measured here; $\Delta E = 1$ corresponding approximately to the just-noticeable difference.
- **HSV/HSI**: Hue Saturation Value/Intensity. Useful for hue manipulation and assessment tasks.
- **CIEDE2000**: an improved colour difference formula that better matches human perception compared to basic ΔE .

2.1.5 CVD Assessment Tests

Ishihara Plates (1917) is the most widely used screening tool [2]. Each plate shows a number hidden within a pattern of coloured dots; viewers with CVD either see a different number or none at all.

Farnsworth Munsell 100 Hue Test (FM100) uses 85 coloured caps distributed across four hue gradients [2]. Participants arrange them in order of hue progression. The test detects all CVD types and provides a severity score, but takes considerable time to complete.

Farnsworth D 15 is a shortened 15 cap version. Less sensitive to mild CVD, but faster.

2.1.6 CVD Simulation Models

Brettel et al. (1997) [8] developed one of the earliest computerized CVD simulations by projecting colours onto confusion lines in LMS space. It is accurate for dichromacy but does not handle anomalous trichromacy.

Machado, Oliveira, and Fernandes (2009) [7] extended the model to support both dichromacy and anomalous trichromacy using a severity parameter $s \in [0, 1]$ where $s = 0$ is normal vision and $s = 1$ is complete dichromacy. Given an input colour vector $\mathbf{c}_{orig} = [R, G, B]^T$, the simulated colour is:

$$\mathbf{c}_{sim} = M_{CVD}(s) \cdot \mathbf{c}_{orig} \quad (2.1)$$

where $M_{CVD}(s)$ is a 3×3 transformation matrix parameterized by CVD type and severity. This model is implemented in Photoshop, Coblis, and professional simulation tools, and is used as the foundation for the proposed system.

2.2 Literature Review

2.2.1 Similar Applications

Several existing tools attempt to address CVD accessibility in digital environments, though each approaches the problem differently and with different trade offs. Understanding what these tools do, and where they fall short, is what shapes the design of our system.

Operating System Built in Filters

Windows, macOS, iOS, and Android all ship with built in colour filter modes [9]. These apply fixed Daltonization transforms at the OS level across the entire screen. The filters were built for specific CVD types at maximum severity, so they offer no adjustment for severity level or individual perceptual variation. Because they apply system wide, they cannot be toggled selectively per application or content type. A user with mild protanomaly (30% severity) gets the same filter as someone with complete protanopia [10].

Dalton Chrome Extension

Dalton is a browser extension that applies fixed recolouring filters with saturation adjustment to web content. Casma Rubio et al. [10] note that it covers dichromacy and some subtypes but does not handle anomalous trichromacy which accounts for the majority of real world CVD cases. All filters are static, there is no user profiling,

and the extension only works inside the browser; it has no effect on any application running outside it.

Chromatic Vision Simulator

Chromatic Vision Simulator is a mobile application that lets users view the world through their camera with CVD perception simulated in real time [10]. It is a tool for designers and non CVD users curious about how colour blind people see not a correction tool for people who actually have CVD. Geddes et al. [4] point out in their 2025 scoping review that simulation only tools are increasingly problematic because they substitute an approximation of CVD experience for the real thing, which can mislead designers and sidestep the actual accessibility problems.

Inclusive Hue (Casma Rubio et al., 2024)

Inclusive Hue is a mobile application that applies recolouring filters using HSV colour space transformation and segmentation algorithms [10]. For dichromats, the pipeline applies subtype-specific RGB transformation matrices (protanopia, deuteranopia, tritanopia), converts to HSI to adjust hue and saturation independently, then converts back to RGB. Anomalous trichromacy is handled through LMS spectral response matrices parameterised per subtype. Monochromacy gets a luminance-preserving grayscale conversion using the NTSC standard, with contrast between colour clusters maximised via the Fletcher-Reeves conjugate gradient method.

The system works on captured images, not on live screen output, and the authors acknowledge this openly. No user testing was conducted; the evaluation was based entirely on heuristic expert review. The authors list real time screen correction and personalised profiles as future work. As it stands, the user must know their own CVD type and select the right filter manually which again assumes a clinical diagnosis that most people with CVD do not have.

Computational Trichromacy Reconstruction (Zhu et al., 2024)

Presented at UIST 2024, this system takes a different angle on the problem entirely [11]. Rather than recolouring content automatically, it lets dichromats interactively induce controlled colour shifts by swiping on a smartphone AR interface. Colours that would normally appear identical to the user shift along different trajectories with each swipe; by observing these shifts and combining them with their existing colour perception, users can reconstruct a functional three dimensional colour space and learn to identify colours they previously confused. The system was tested in real world scenarios, interpreting artwork, building with Lego blocks, and users reported being able to

recognise colours they otherwise could not.

The limitation in this paper is that it is a learning and recognition tool, not a passive correction system. It needs active interaction and a training period, which makes it unsuitable for continuous screen correction during everyday computing. It only targets dichromats and does not address anomalous trichromacy.

Hue4U (Qin et al., 2025)

Hue4U is the most directly comparable prior system [2]. It is a real time, personalised CVD correction system built in Unity on a Meta Quest 3. The assessment module uses an adapted FM 100 Hue Test in AR, reduced from 85 to 48 movable caps to limit user fatigue while preserving diagnostic accuracy. Caps are interpolated in HSV colour space rather than RGB to maintain perceptual consistency across hue gradients. Using per row arrangement errors, the system classifies CVD type and estimates severity on a $[0, 1]$ scale through three cumulative error scores (E_{protan} , E_{deutan} , E_{tritan}).

The correction pipeline follows a three stage Daltonization approach using the Machado Oliveira Fernandes model [7]. First, it simulates CVD perception using a severity parameterised matrix $M_{\text{CVD}}(s)$. Then it computes the colour error $\mathbf{e} = \mathbf{c}_{\text{orig}}\mathbf{c}_{\text{sim}}$. Finally, it redistributes this error via a correction matrix C_{type} to produce $\mathbf{c}_{\text{corr}} = \mathbf{c}_{\text{orig}} + C_{\text{type}} \cdot \mathbf{e}$. The correction runs per pixel in real time through a custom Unity URP shader at roughly 150 ms latency.

A user study with 10 participants showed statistically significant improvements on the Ishihara plate test: a mean difference of 8.2 plates out of 25 across all participants ($d = 2.46$, $p < .001$) and 9.29 plates for the confirmed CVD subgroup ($d = 1.45$) [2]. Notably, three participants who scored normally on a desktop FM 100 test still failed several Ishihara plates without correction, which suggests that using multiple assessment methods picks up mild anomalies that a single test can miss.

The main limitation is hardware: the system requires a Meta Quest 3 headset and only corrects AR rendered virtual content, not general screen output. The DVAL system was designed to remove exactly this constraint.

The Broader Picture

Geddes et al.’s 2025 scoping review of 113 CVD accessibility papers found that 72% of design tools relied on simulation as their primary feature, while fewer than 15% offered colour scheme generation or any form of automated colour adaptation [4]. The review identifies over reliance on simulation and the absence of personalisation as the two most persistent gaps in the field.

What none of the tools above do is combine all three properties at once: no specialist hardware, real time correction across arbitrary screen content, and severity conditioned correction personalised through in browser profiling without prior clinical diagnosis. That combination is the core design goal of DVAL.

2.2.2 Related Research

The academic literature on CVD accessibility spans four main groups, reviewed in the following.

2.2.2.1 *Classical Daltonization and Fixed Filter Methods*

The foundation of computational CVD correction was established by Viénot et al. [12] who introduced the concept of digital video colour maps for dichromats: the original daltonization. Brettel et al. [8] provided the physiological basis by developing a computerized simulation of dichromatic colour appearance using confusion lines in LMS space. The Machado et al. [7] model significantly extended this by supporting anomalous trichromacy with a parameterized severity dimension, and this model has since become the standard in most professional tools including Photoshop.

A survey of colour-changing algorithms by Ribeiro and Gomes (2019) looked at all major approaches for colour-blind people from the past 30 years of research. This survey found that most people are still using a method that is meant for people with dichromacy which is a type of colour blindness and it is a fixed way of changing colours. There is not work on helping people with anomalous trichromacy, which is another type of colour blindness or making things personal for each user. [3].

The big problem with these methods is that they all apply fixed, static transformations. As Geddes et al. note [3], most colour correction tools are validated against tests like the Ishihara plate — a diagnostic measure, not a usability one. That focus reveals a broader bias in the field: the design goal has been correcting for blindness rather than improving everyday usability for the much larger population with partial deficiency. Users with CVD have made this frustration clear. One-size-fits-all filters are widely disliked, and a recurring complaint is that the tools do not reflect what people actually struggle with. What users want is personalisation — algorithms tuned to their specific type and severity, not a generic approximation of what a dichromat sees.[3].

2.2.2.2 *Deep Learning Based Recolouring*

More recent work has applied neural networks to the recolouring problem. Chen et al. [13] proposed a DNN based recolouring method using Swin Transformer blocks, achieving near real time performance competitive with state of the art quality. The

authors note that their unsupervised training approach, as well as the dataset of images with confusing colours for CVD individuals, can be used to develop other image synthesis technologies. However, a key limitation they acknowledge is that the system targets dichromacy as the compensation target, and recolouring results may be unsuitable for individuals with lower degrees of CVD [13].

At ICCV 2023, Jiang et al. [14] presented CVD GAN, which achieves personalized image generation by disentangling and controlling a severity parameter δ_s parameterized by the shift in peak cone sensitivity. The GAN based approach avoids the non CVD oriented limitations of combining standard generation models with post processing recolouring, achieving personalized generation at the content level. However, the system targets static image generation and not screen level real time rendering.

At the Frontiers journal level, Pendhari et al. [15] demonstrated autoencoder based enhancement of Ishihara plates with a One vs All SVM classification model achieving 99.7% accuracy on CVD type detection, and surveys with 15 diagnosed students confirmed substantial perceptual improvement. The work, while focusing on educational images, demonstrates the viability of ML based CVD type classification from plate response patterns: a technique directly applicable to the profiling module in the proposed system.

2.2.2.3 Personalized and Degree Adaptive Approaches

Zhu et al. [5] represents the most important prior work on personalized recolouring. They introduced a degree adaptable recolouring algorithm that recolours images by minimizing an objective function constrained by contrast enhancement and naturalness preservation, adapting to different degrees of CVD through a severity parameter. Their evaluation with 14 volunteers with varying degrees of CVD showed that the personalized method outperforms state of the art methods. The critical limitation is computational cost: the optimization based approach requires approximately 30 seconds per image [2], making it entirely unsuitable for real time applications.

Qin et al. [2] addressed the real time limitation within an AR context by replacing the optimization with a pre computed Machado matrix look up conditioned on the user's CVD profile. Their system achieves approximately 150ms end to end pipeline latency on Meta Quest 3. Critically, the profiling uses a modified 48 cap FM100 test rather than the full 85 cap test, and the classification computes cumulative positional errors

per CVD type:

$$E_{protan} = E_{row_1} + E_{row_4} \quad (2.2)$$

$$E_{deutan} = E_{row_1} + E_{row_2} \quad (2.3)$$

$$E_{tritan} = E_{row_3} + E_{row_4} \quad (2.4)$$

$$\text{CVD Type} = \arg \max\{E_{protan}, E_{deutan}, E_{tritan}\} \quad (2.5)$$

This profiling approach is adapted in the proposed system for the browser based assessment module.

2.2.2.4 UI and Screen Accessibility Studies

Jamil and Denes [16] investigated colour blind UI accessibility through simulated interfaces, finding a positive correlation between perceived aesthetics and functionality, and importantly, that high contrast modes can reduce both. This supports the need for personalised rather than maximally contrasted solutions.

Sajek et al. [6] conducted the most recent and comprehensive study of colour scheme perception across CVD types, surveying 18 CVD participants across 12 colour scheme variants of a healthcare web application. Their analysis using two way repeated measure ANOVA confirmed that both primary colour choice and colour scheme type have statistically significant effects on perceived accessibility ($p < 0.005$ in all cases), and that colour accessibility depends not only on contrast but on the specific combination of colours for a given CVD type. Triadic colour schemes with appropriate primary colours achieved the highest weighted accessibility scores across all CVD types.

The scoping review by Geddes et al. [4] analysed 113 academic papers on CVD accessible design methods. The review found that the most common design tool feature is CVD simulation (72% of design tools), with solution generation (accessible palette suggestion) appearing in only 13% of tools. Of the guideline papers, only 10% included user studies with actual CVD participants. The review identified ten research opportunities for the field, most relevantly including developing design tools that actively assist in colour selection (not just identification of problems), and improving the connection between real time tools and personalised user colour models.

2.3 Gap Analysis

A synthesis of the literature reveals that existing work has addressed the following aspects of CVD accessibility:

- Fixed filter daltonization for dichromacy : thoroughly solved by Viénot [12], Brettel [8], and Machado [7]. This literature provides the mathematical foundation for the proposed system but is not in itself a research contribution.
- Deep learning recolouring of static images: well covered by Chen et al. [13] and Jiang et al. [14]. These serve as quality and performance baselines.
- Anomalous trichromacy modelling with severity parameter: partially addressed by Zhu et al. [5] but computationally too expensive for real time.
- Real time personalized correction: demonstrated by Hue4U [2] but requiring AR headset hardware and limited to AR content.

The following gaps remain unaddressed in the existing literature:

Table 2.1: Gap Analysis: Prior Work vs. Proposed Dynamic Visual Accessibility Layer

Research Gap	Best Work	Prior Work	Limitation of Prior Work	Status	This Work
Fixed filter daltonization for dichromacy	Viénot Brettel Machado [7]	[12]; [8];	Thoroughly solved. Provides mathematical foundation but is not a research contribution.	Closed	Used as baseline
DL based re-colouring of static images	Chen et al. [13]; Jiang et al. [14]		Targets static images only. No real time screen rendering. Most methods still target dichromacy.	Closed	Used as quality baseline
Degree adaptive correction for anomalous trichromacy	Zhu et al. [5] : optimization based severity parameter δ		~30 seconds per image; unsuitable for real time use. No screen layer deployment.	Partial	Extended to real time
Real time personalized correction without specialist hardware	Hue4U [2]: AR headset, no prior diagnosis		Requires Meta Quest 3 AR headset. Corrects AR content only; unusable on standard screens.	Partial	Primary contribution
End to end pipeline: profiling $\rightarrow$ model $\rightarrow$ screen overlay	No existing system combines all three stages as a unified cross application software layer [4]		OS filters are static; AR systems require hardware; static image systems are not real time. No end to end integration exists.	Open	Primary contribution
User driven adaptive profile refinement	No existing screen accessibility tool supports in use feedback based profile refinement [3]		CVD users have nuanced self knowledge of their vision [3] but no tool exploits this.	Open	Secondary contribution

The proposed Dynamic Visual Accessibility Layer directly targets Gaps 1 (Fixed filter daltonization for dichromacy) and 2 (DL based re-colouring of static images) as primary contributions, with Gaps 3 (Degree adaptive correction for anomalous trichromacy) and 4 (Real time personalized correction without specialist hardware) addressed as secondary contributions. As stated in the novelty framing: *while prior work has addressed personalized recolouring of static images [5] and real time AR based correction [2], no existing system integrates automated severity profiling with a learned colour transformation model deployed as a cross application screen accessibility layer. This work addresses that gap.*

2.4 Summary

This chapter established the physiological and computational foundations of CVD, including cone cell biology, the two dimensional classification of CVD types and severities, colour confusion lines, the key colour spaces used in modelling, and the standard assessment tests. The literature review surveyed four clusters of prior work: classical daltonization, DL based recolouring, personalized and degree adaptive approaches, and UI/screen accessibility studies, with direct engagement with all uploaded reference papers. The gap analysis identified four specific unaddressed problems in the literature that this project targets, with the primary novelty lying in the end to end personalised real time screen correction pipeline described in the following chapter.

Chapter 3

Project Design

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

Functional Requirements

- FR1. CVD Assessment:** The system shall present an interactive hue ordering assessment to the user and derive a CVD type classification (protan, deutan, tritan, or normal) and a severity score $\delta \in [0, 1]$ from the user's responses.
- FR2. User Profile Storage:** The system shall store the user's CVD profile (type and severity) persistently in browser local storage for use in subsequent sessions.
- FR3. Real Time Colour Correction:** The system shall apply a severity conditioned colour transformation to all visible screen pixels in real time (target: ≥ 30 FPS) when the correction layer is enabled.
- FR4. Toggle Control:** The user shall be able to enable or disable colour correction at any time via the browser extension popup.
- FR5. Severity Adjustment:** The user shall be able to manually fine tune the severity parameter δ via a slider control after initial profiling.
- FR6. Type Selection Override:** If the automatic assessment gets it wrong, the user can manually select or override their CVD type.
- FR7. Baseline Comparison Mode:** The user shall be able to switch between fixed standards and personalized correction. This is useful for seeing exactly what the system is doing differently.
- FR8. Re-profiling :** The user can retake the CVD assessment at any time to update their profile.

Non-functional Requirements

- NR1. Performance:** The colour transformation pipeline must run at a minimum of 30 FPS on mid-range consumer hardware, with 60 FPS as the target.
- NR2. Naturalness:** Colour corrections shall preserve image naturalness, minimizing

visual artefacts and unnatural saturation shifts of the image that were already accessible to the user.

- NR3. Accuracy:** The CVD type classification module shall achieve at least 85% type classification accuracy on a held out test set of simulated assessment responses.
- NR4. Usability:** The onboarding CVD assessment shall be completable in under 3 minutes by a first time user with no prior colour vision testing experience.
- NR5. Compatibility:** The browser extension shall be compatible with Chromium based browsers (Chrome, Edge) supporting the Manifest V3 WebExtensions API.
- NR6. Privacy:** No user data, screen content, or CVD profile information shall be transmitted to any external server; all processing shall occur locally in the browser.
- NR7. Accessibility:** The system UI (assessment interface and extension popup) shall itself conform to WCAG 2.1 Level AA standards, so that it is operable by CVD users without requiring the correction layer to be active.
- NR8. Maintainability:** The codebase shall be modular with separate modules for assessment, profiling, model inference, and rendering, each independently testable.

3.1.2 Context Diagram

Figure 3.1 shows the Level 0 context diagram for the Dynamic Visual Accessibility Layer system. The system interacts with three external entities: the *CVD User* who provides assessment responses and profile adjustments; the *Browser* which provides the screen rendering pipeline and extension APIs; and the *Pretrained Model* which is a static artefact loaded at initialization.

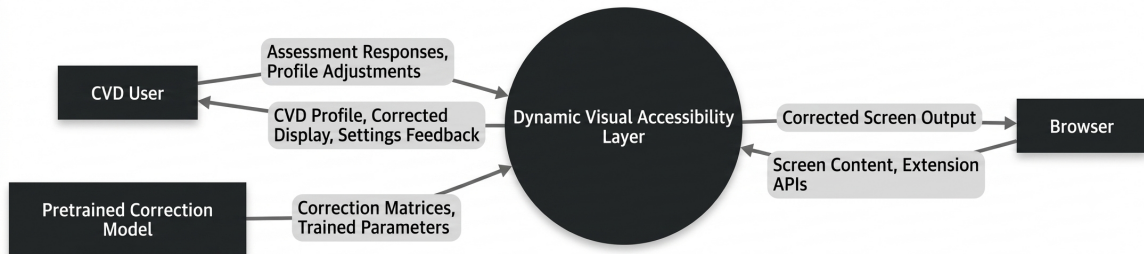


Figure 3.1: Level 0 Context Diagram for the Dynamic Visual Accessibility Layer.

3.1.3 Data Flow Diagram Level 1

The Level 1 DFD (Figure 3.2) decomposes the DVAL system into four primary processes: (1) *Assessment Engine* presents hue ordering tasks and records user responses; (2)

Profile Estimator computes CVD type and severity δ from assessment errors; (3) *Correction Model* applies the severity conditioned colour transformation matrix; and (4) *Screen Renderer* intercepts browser rendering and applies per pixel correction via WebGL shader.

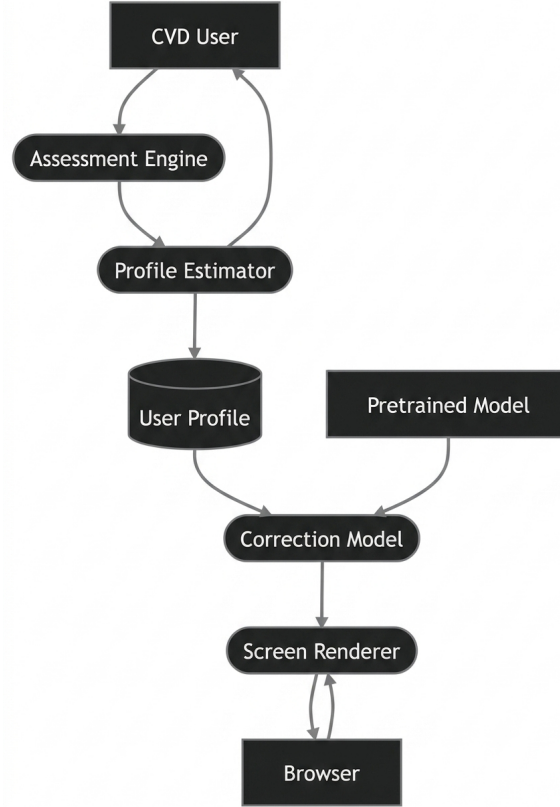


Figure 3.2: Level 1 Data Flow Diagram for the Dynamic Visual Accessibility Layer.

3.1.4 UI Design

The DVAL system has two primary UI components:

Assessment Interface: A browser based page presenting a hue ordering task. The user is shown a row of coloured tiles with two fixed anchor colours at each end, and must drag intermediate tiles into the correct hue gradient order. Four rows are presented, covering hue ranges chosen to differentially affect protan, deutan, and tritan response patterns (following the design in Hue4U [2]). The interface adapts to known accessible colour combinations for the task presentation itself. On completion, results are automatically analysed and the user is shown their estimated CVD type and severity.

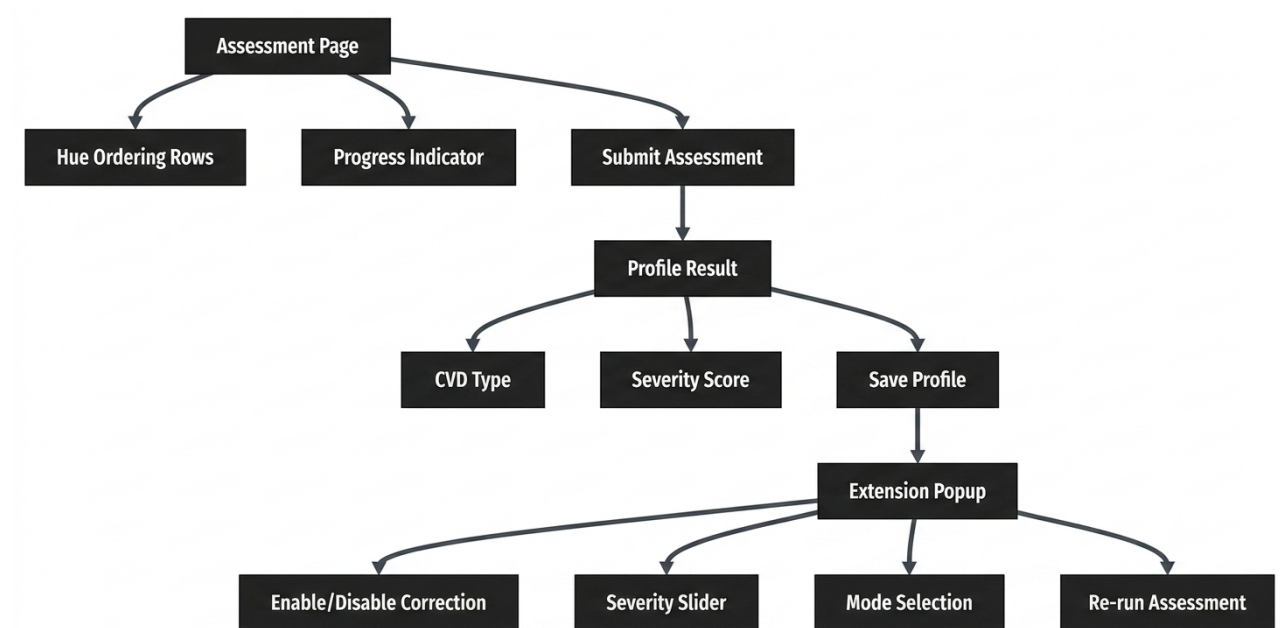


Figure 3.3: CVD assessment interface flow

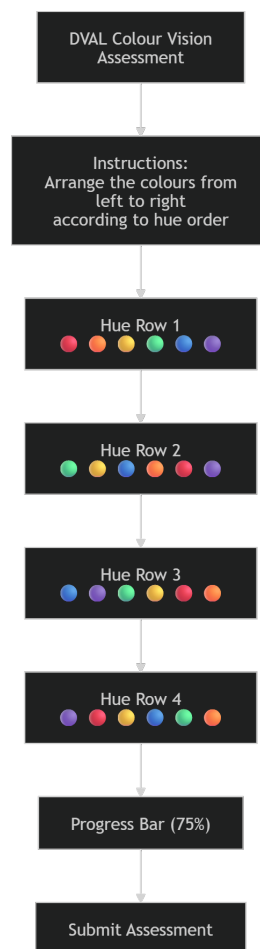


Figure 3.4: Assessment Interface Mockup

Extension Popup: A compact browser extension popup showing: the current CVD profile (type and δ); an on/off toggle for correction; a severity slider for manual fine tuning; a mode selector (personalized / standard / off); and a button to re run the assessment. All controls are accessible via keyboard and are labelled with ARIA attributes.

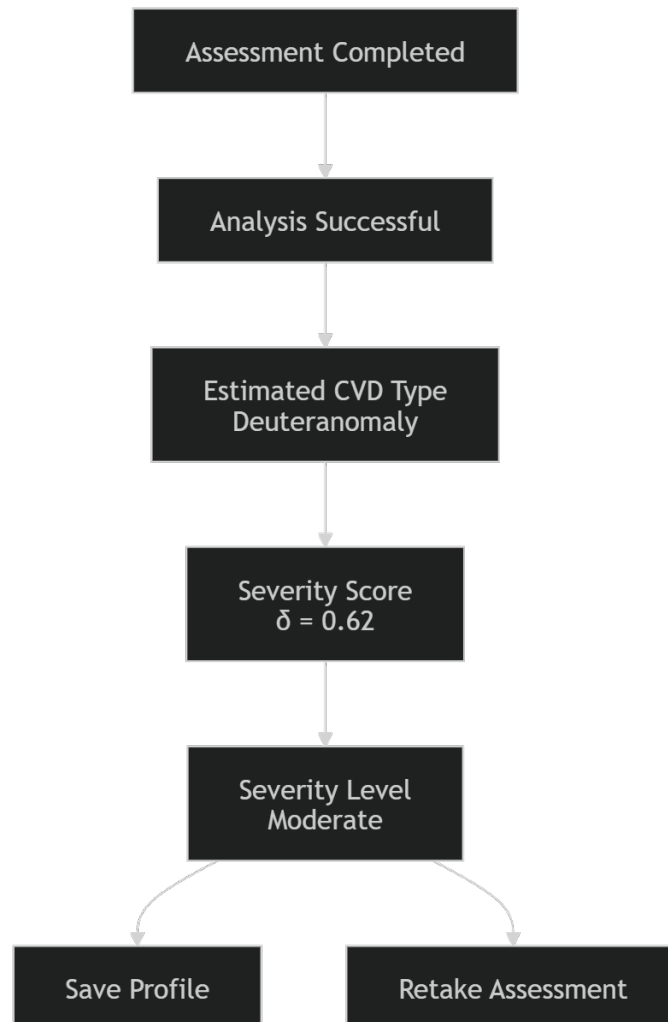


Figure 3.5: Assessment Result Screen

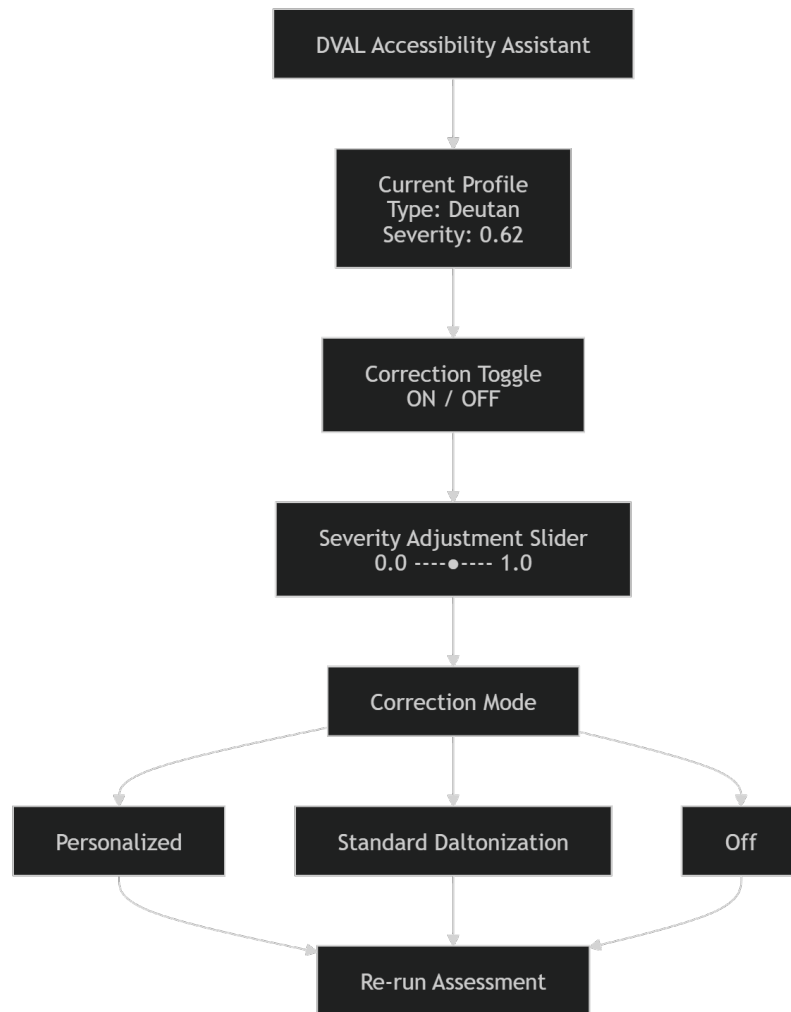


Figure 3.6: Extension control popup.

3.2 Detailed Methodology and Design

3.2.1 Phase 1: Dataset Construction and Model Training

3.2.2 Phase 2: CVD Assessment and Profiling

3.2.3 Phase 3: Real Time Screen Rendering

3.3 Project Plan

Table 3.1 shows the project timeline distributed across the academic semester.

Table 3.1: Project Timeline and Milestones

Task	Start	End	Deliverable
Literature review & gap analysis	Week 1	Week 3	Chapter 2 draft
System architecture design	Week 2	Week 4	Architecture diagram, DFDs
Dataset construction	Week 3	Week 5	CVD simulation image pairs
Model training & validation	Week 5	Week 8	Trained correction model
Assessment module implementation	Week 6	Week 9	Browser based assessment UI
Browser extension development	Week 8	Week 12	Working extension (Chrome)
Integration testing	Week 11	Week 13	Integration test report
User study	Week 13	Week 14	User study data
Results analysis & write up	Week 14	Week 16	Full thesis draft
Supervisor review & revision	Week 16	Week 17	Final thesis
Submission	Week 18	Week 18	Submitted thesis

3.4 Task Allocation

Table 3.2: Task Allocation

Task	Assigned To	Notes
Dataset generation & pre-processing	[Kafi]	Python, NumPy
Model training & evaluation	[Kafi, Sanjida]	PyTorch
Assessment module (frontend)	[Rafi]	HTML, JS, CSS
Browser extension (content script)	[Zobayer]	WebGL, JS
Extension popup UI	[Mohai]	HTML, CSS, JS
User study design & conduction	[Rafi]	
Thesis writing & documentation	[Sanjida, Kafi]	LaTeX

3.5 Summary

This chapter presented the complete design of the Dynamic Visual Accessibility Layer. Functional and nonfunctional requirements were specified, followed by the system context and Level 1 data flow diagrams. The UI design for both the assessment interface and extension popup were outlined. The detailed methodology described all three phases of the system: dataset construction and model training using Machado simulation matrices, the hue ordering assessment with mathematical derivation of CVD type and severity, and the real time WebGL shader pipeline for browser level screen correction. The project plan and task allocation establish the execution roadmap for the implementation described in the following chapter.

Chapter 4

Implementation and Results

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

5.1 Compliance with the Standards

5.1.1 Software Standards

Web Content Accessibility Guidelines (WCAG) 2.1. The DVAL system itself complies with WCAG 2.1 Level AA, the internationally recognized standard for web content accessibility [4]. This includes ensuring all UI elements in the assessment interface and extension popup have sufficient contrast ratios ($\geq 4.5:1$ for normal text, $\geq 3:1$ for large text), all interactive elements are keyboard operable, and all images and icons have text alternatives. The system is designed to be usable by CVD users even before any correction filter is activated.

WebExtensions API / Manifest V3. The browser extension follows the Chrome Manifest V3 specification, which is the current standard for Chromium based browser extensions. This includes using the service worker background script model, declarative content rules, and the permissions model defined in the Manifest V3 API documentation.

OpenGL ES Shading Language (GLSL). The pixel shader pipeline uses WebGL 2.0 with GLSL ES 3.00, following the Khronos Group OpenGL ES 3.0 specification. Shader code adheres to precision qualifiers, texture sampling standards, and clamp operations required for correct cross platform colour output.

IEEE 754 Floating Point Standard. All colour computation in the shader pipeline uses 32 bit floating point arithmetic conforming to IEEE 754, ensuring deterministic colour correction results across platforms.

ISO/IEC 13211 (Colour Measurement). Colour difference calculations use the CIEDE2000 formula (ΔE_{00}), which is the current industry standard for measuring perceptual colour difference, superseding the older CIE76 formula used in earlier CVD research [6].

5.1.2 Hardware Standards

Display Colour Accuracy. All testing was done on monitors calibrated to sRGB, which is the colour space the Machado model assumes [7]. On wide-gamut displays

like DCI-P3 panels, results may vary sRGB mode is recommended for evaluation.

GPU Compute Requirements. The system needs a GPU supporting WebGL 2.0 (OpenGL ES 3.0 equivalent), which covers every consumer GPU made after 2012. Nothing beyond standard fragment shading is required such as no compute shaders, no specialised hardware.

5.1.3 Communication Standards

Privacy by Design: All processing happens locally on the user’s device. No screen content, CVD profile, or assessment results are sent over any network. This aligns with GDPR Article 25 (data protection by design) and Bangladesh’s Digital Security Act 2018 provisions on personal data.

Extension Communication: The background service worker, content script, and pop-up communicate through the Chrome Extension Message Passing API. A sandboxed IPC mechanism that enforces least-privilege access. Content scripts see only what the background script explicitly exposes, nothing more.

5.2 Design Constraints

5.3 Cost Analysis

Table 5.1: Estimated Development Cost Analysis

Item	Tool/Resource	Cost	Notes
Model training	Google Colab	BDT 0	Colab free tier
Source image dataset	MS-COCO 2017	BDT 0	Freely available
Development IDE	VS Code	BDT 0	Open source
Version control	GitHub	BDT 0	Free tier
Browser testing	Chrome, Edge	BDT 0	Free
Printed study materials	–	BDT 450	–
Poster printing (A1, colour)	–	BDT 800	–
Thesis printing & binding (5 copies)	–	BDT 2500	–
Hardware	PC	BDT 80,000	–
User study materials	Printed consent forms	BDT 500	–
Human hours (development)	5 members × 240 hrs × BDT 100/hr	BDT 120,000	Student labour
Participant refreshments	–	BDT 5000	–
Total estimated cost	–	≈ BDT 209,250	–

Compared to existing CVD solutions such as EnChroma glasses (\$300–400 USD), or AR headset systems (\$500+ USD), our software system is much more affordable to develop, making it significantly more accessible for the target population.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

This project addresses a complex engineering problem under the Washington Accord’s graduate attribute framework. Four characteristics justify that classification:

Depth of knowledge required: The solution requires knowledge across computer vision, colour science (photoreceptor physiology, CIE colour spaces, perceptual models), machine learning (training and evaluation), web engineering (browser extensions,

WebGL shaders), and human-computer interaction (user study design, accessibility standards).

Range of conflicting requirements: The system must simultaneously achieve real time performance that requires computational efficiency, naturalness preservation limits correction aggressiveness, higher correction quality favours heavier models, and platform compatibility restricts the available APIs. No single design choice satisfies all four trade-offs.

Depth beyond routine methods: The core pipeline: severity conditioned Daltonization delivered as a real-time browser screen layer has no prior precedent in the literature as established in Section 2.3. The system required original engineering design, not adaptation of a known method.

Involves multiple components: The system integrates a physiological colour model, a browser extension framework, a GPU shader pipeline, an interactive assessment interface, and a user profile persistence system.

5.4.2 Engineering Activities

The project required the following engineering activities:

Design: Architecture design of the three phase system; DFD and context diagram; UI wireframing; shader pipeline design.

Analysis: Literature analysis to identify the research gap; mathematical analysis of the Daltonization equations; statistical analysis of user study results.

Implementation: Python based dataset construction and model training; JavaScript/WebGL extension development; HTML/CSS assessment UI.

Testing and Evaluation: Unit testing; performance profiling; user study design, execution, and statistical analysis.

5.5 Summary

This chapter addressed the compliance of the Dynamic Visual Accessibility Layer with relevant software, hardware, and communication standards, including WCAG 2.1, WebExtensions Manifest V3, GLSL ES 3.0, and IEEE 754. Seven categories of design constraints were examined: economic, environmental, ethical, health and safety, social, political, and sustainability. A cost analysis established the near zero deployment cost advantage of the software only approach relative to hardware alternatives. The complex engineering problem characterization confirmed the graduate level depth and breadth of the technical problem addressed by this project.

Bibliography

- [1] M. Simunovic, “Colour vision deficiency,” *Eye*, vol. 24, no. 5, pp. 747–755, 2009.
- [2] J. Qin, S. Checherin, Y. Li, B.-J. van der Zwaag, and Ö. Durmaz Incel, “Hue4U: Real-time personalized color correction in augmented reality,” in *Proceedings of the International Workshop on Sensor-Based Activity Recognition and Artificial Intelligence (iWOAR 2025)*, ser. Lecture Notes in Computer Science, vol. 16292. Springer, 2026, arXiv:2509.06776v3.
- [3] C. Geddes, “Designing digital content to accommodate for colour vision deficiency,” in *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA ’23)*, Hamburg, Germany, 2023, pp. 1–5.
- [4] C. Geddes, E. C. Eggertson, J. Sutton, and G. W. Tigwell, “Designing for colour vision deficiency: A scoping review of resources that support designers in choosing accessible colours,” in *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’25)*, Denver, CO, USA, 2025.
- [5] Z. Zhu, M. Toyoura, K. Go, K. Kashiwagi, I. Fujishiro, T.-T. Wong, and X. Mao, “Personalized image recoloring for color vision deficiency compensation,” *IEEE Transactions on Multimedia*, vol. 24, pp. 1721–1734, 2022.
- [6] D. Sajek, O. Korotenko, and T. Kyrychok, “Research on the accessibility of different colour schemes for web resources for people with colour blindness,” *Journal of Imaging*, vol. 11, no. 8, p. 268, 2025.
- [7] G. M. Machado, M. M. Oliveira, and L. A. F. Fernandes, “A physiologically-based model for simulation of color vision deficiency,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 15, no. 6, pp. 1291–1298, 2009.
- [8] H. Brettel, F. Viénot, and J. D. Mollon, “Computerized simulation of color appearance for dichromats,” *Journal of the Optical Society of America A*, vol. 14, no. 10, pp. 2647–2655, 1997.
- [9] Microsoft Corporation, “Use color filters in Windows,” Microsoft Support, 2023. [Online]. Available: <https://support.microsoft.com/en-us/windows/use-color-filters-in-windows>
- [10] S. N. Casma Rubio, J. S. Bautista Villaverde, and I. C. Bautista Fuentes, “Filters and recoloring algorithms to improve mobile accessibility for users with color

- blindness,” in *Proceedings of the 2024 2nd International Conference on Electronics, Computers and Communication Technology (CECCT 2024)*, Chengdu, China, 2024, pp. 227–233.
- [11] Y. Zhu, E. Chen, C. Hascup, Y. Yan, and G. Sharma, “Computational trichromacy reconstruction: Empowering the color-vision deficient to recognize colors using augmented reality,” in *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, ser. UIST ’24. ACM, 2024, pp. 1–17.
- [12] F. Viénot, H. Brettel, and J. D. Mollon, “Digital video colourmaps for checking the legibility of displays by dichromats,” *Color Research & Application*, vol. 24, no. 4, pp. 243–252, 1999.
- [13] L. Chen, Z. Zhu, W. Huang, K. Go, X. Chen, and X. Mao, “Image recoloring for color vision deficiency compensation using Swin transformer,” *Neural Computing and Applications*, vol. 36, pp. 6051–6066, 2024.
- [14] S. Jiang, D. Liu, D. Li, and C. Xu, “Personalized image generation for color vision deficiency population,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2023)*, 2023, pp. 22 571–22 580.
- [15] S. Pendhari *et al.*, “Enhancing Ishihara and educational images using machine learning: Toward accessible learning for colorblind individuals,” *Frontiers in Artificial Intelligence*, 2025.
- [16] A. Jamil and G. Denes, “Investigating color-blind user-interface accessibility via simulated interfaces,” *Computers*, vol. 13, no. 2, p. 53, 2024.